

100 Best Python List Practice Programs (College + Placement)

This roadmap goes from **Beginner** → **Intermediate** → **Advanced**, covering the most common list questions asked in college practicals and coding interviews.

Beginner Level

1. Print all elements of a list
 2. Find the length of a list
 3. Sum of all elements
 4. Find the largest element
 5. Find the smallest element
 6. Find the average of elements
 7. Count even and odd numbers
 8. Count positive and negative numbers
 9. Find maximum and minimum in one traversal
 10. Reverse a list
 11. Copy a list
 12. Check if a list is empty
 13. Access first and last element
 14. Replace an element
 15. Insert an element at a specific position
 16. Delete an element
 17. Remove all occurrences of an element
 18. Clear a list
 19. Concatenate two lists
 20. Repeat a list (`*` operator)
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Easy-Intermediate

21. Find second largest element
 22. Find second smallest element
 23. Find third largest element
 24. Find index of an element
 25. Count occurrence of an element
 26. Find frequency of each element
 27. Find the most frequent element
 28. Find the least frequent element
 29. Find all duplicate elements
 30. Find first duplicate element
 31. Remove duplicate elements
 32. Remove consecutive duplicates
 33. Find unique elements
 34. Sort a list in ascending order
 35. Sort a list in descending order
 36. Sort without using `sort()`
 37. Check if list is sorted
 38. Merge two sorted lists
 39. Merge two unsorted lists
 40. Swap two elements
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Intermediate

41. Swap first and last element
 42. Rotate list left by one
 43. Rotate list right by one
 44. Rotate list by K positions
 45. Move all zeros to the end
 46. Move all zeros to the beginning
 47. Move negative numbers to the beginning
 48. Move positive numbers to the end
 49. Separate even and odd numbers
 50. Reverse only even numbers
 51. Reverse only odd numbers
 52. Reverse a sublist
 53. Find common elements of two lists
 54. Find union of two lists
 55. Find intersection of two lists
 56. Find difference between two lists
 57. Find symmetric difference
 58. Remove common elements
 59. Find missing number (1 to N)
 60. Find missing numbers in a sequence
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Intermediate-Advanced

61. Find pairs with a given sum
62. Find triplets with a given sum
63. Find pairs with a given difference
64. Find product of all elements
65. Find prefix sum
66. Find suffix sum
67. Running sum of a list
68. Running maximum
69. Running minimum
70. Find equilibrium index
71. Kadane's Algorithm (Maximum subarray sum)
72. Minimum subarray sum
73. Longest increasing subarray
74. Longest decreasing subarray
75. Maximum product subarray

Advanced

- 76. Check if list is palindrome
 - 77. Find Kth largest element
 - 78. Find Kth smallest element
 - 79. Find median of a list
 - 80. Find mode of a list
 - 81. Find all leaders in a list
 - 82. Find majority element ($> n/2$)
 - 83. Find elements occurring more than $n/3$ times
 - 84. Longest consecutive sequence
 - 85. Find peak element
 - 86. Find first repeating element
 - 87. Find first non-repeating element
 - 88. Find closest pair
 - 89. Find smallest missing positive number
 - 90. Rearrange list in alternating positive and negative order
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Interview Level

- 91. Flatten a nested list
 - 92. Spiral traversal of a matrix
 - 93. Rotate a matrix by 90°
 - 94. Transpose a matrix
 - 95. Search in a sorted matrix
 - 96. Merge overlapping intervals
 - 97. Find inversion count
 - 98. Dutch National Flag (Sort 0s, 1s, 2s)
 - 99. Trapping Rain Water
 - 100. Next Greater Element
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Python Built-in Practice

- 101. `append()`
 - 102. `extend()`
 - 103. `insert()`
 - 104. `remove()`
 - 105. `pop()`
 - 106. `clear()`
 - 107. `index()`
 - 108. `count()`
 - 109. `sort()`
 - 110. `reverse()`
 - 111. `copy()`
 - 112. List slicing
 - 113. List comprehension
 - 114. Nested lists
 - 115. Zip two lists
 - 116. Enumerate a list
 - 117. Any and All
 - 118. Map with list
 - 119. Filter with list
 - 120. Convert list ↔ tuple ↔ set ↔ string
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 **If you complete these 120 programs, you'll cover almost every important Python list concept used in:**

-  College practicals and viva
-  Semester exams
-  Placement coding rounds
-  HackerRank
-  LeetCode (easy to medium)
-  Infosys, TCS, Wipro, Accenture, Capgemini, Cognizant, Deloitte, etc.